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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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One-Pass vs Multi-Pass Compiler

CSA1407– COMPILER DESIGN

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UNDER THE GUIDANCE

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What is a Compiler?



Compiler

- Converts source code into machine code.
- Checks for syntax and semantic errors.
- Generates an executable program.
- Acts as a translator between the programmer and the computer.
- Performs different phases such as lexical analysis, syntax analysis, semantic analysis, code optimization, and code generation.

Types of Compiler

- **One-Pass Compiler** – Reads the source code only once.
- **Multi-Pass Compiler** – Reads the source code multiple times for better optimization.



ONE-PASS COMPILER



One-Pass Compiler

- Reads the source code only once.
- Fast execution.
- Uses less memory.
- Suitable for simple programming languages.
- Simple and easy to implement.
- Produces machine code during a single scan.

Advantages

- Faster compilation time.
- Requires less memory.

Disadvantages

- Limited error detection.
- Cannot perform advanced code optimization.



MULTI-PASS COMPILER



Multi-Pass Compiler

- Reads the program multiple times.
- Better error detection.
- Produces optimized code.
- Used for complex programming languages.
- Generates high-quality and efficient machine code.

Advantages

- Better error detection and reporting.
- Produces optimized and efficient code.

Disadvantages

- Slower compilation time.
- Requires more memory.

